Inventory and Stock Management System (ISMS) – Data Flow Diagrams

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Aim

To model the flow of data through the Inventory and Stock Management System (ISMS) using Data Flow Diagrams at Level 0 (context), Level 1 and Level 2, drawn in Gane-Sarson notation and derived strictly from the system's class diagram and activity diagram.

The diagrams use four symbols:

- **Process** – a rounded rectangle with the process number in the top band; it transforms incoming data into outgoing data.
- **Data store** – an open-ended rectangle with its ID (D1, D2 …) in the left compartment; each store corresponds to a class in the class diagram.
- **External entity** – a plain rectangle; an actor or external service from the activity diagram that sends or receives data.
- **Data flow** – a labelled arrow showing the data moving between the other three.

DFD Level 0 – Context Diagram

The whole ISMS is shown as a single process (0) exchanging data with eleven external entities: the four user roles from the activity diagram's swimlanes (Administrator, Manager, Sales Personnel, Warehouse Staff), the Supplier class, and the six services in the External Services lane (Marketplace Webhook, Payment Gateway, Email / SMS Gateway, Speech-to-Text, Text-to-Speech and ML Forecasting).

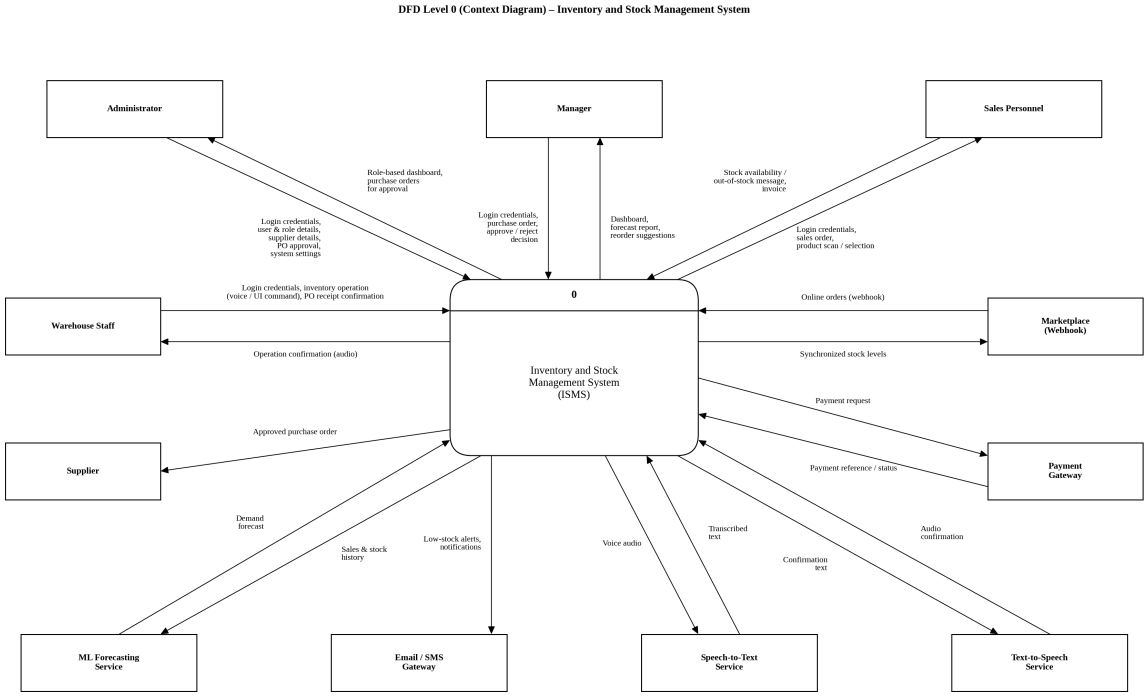


Figure 1: DFD Level 0 – Context Diagram

DFD Level 1

Process 0 is broken into seven processes, one per responsibility in the activity diagram, and eight data stores, one per class in the class diagram. Duplicated entities and stores (repeated to avoid crossing lines) carry the Gane-Sarson duplicate mark: a diagonal line in the corner of an entity, an extra vertical line in a store's ID box.

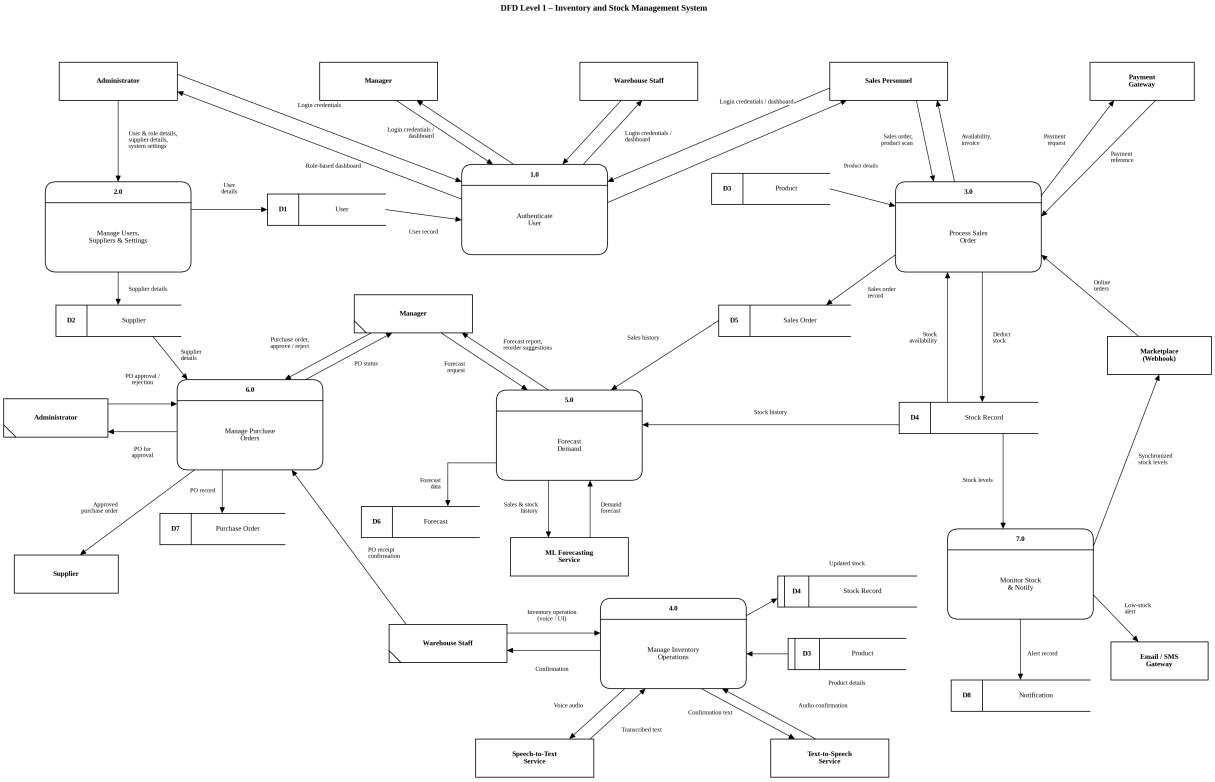


Figure 2: DFD Level 1

Process	**Source in the activity / class diagram**
1.0 Authenticate User	Validate Login Credentials → Identify User Role → Display Role-Based Dashboard; User.login()
2.0 Manage Users, Suppliers & Settings	Administrator lane: Manage Users / Roles, Manage Suppliers, Configure System Settings
3.0 Process Sales Order	Sales Personnel lane: Create Sales Order to Update Sales Order; Marketplace Webhook; Payment Gateway
4.0 Manage Inventory Operations	Warehouse Staff lane: Select Inventory Operation to Update Stock Record; Speech-to-Text and Text-to-Speech services
5.0 Forecast Demand	Manager lane: Generate Forecast Report, Review Reorder Suggestions; ML Forecasting Service; Forecast class
6.0 Manage Purchase Orders	Create Purchase Order, Approve / Reject (Manager and Administrator); Supplier.fulfills, WarehouseStaff.receive_po_confirm()
7.0 Monitor Stock & Notify	Synchronize Across Channels, Check Reorder Levels, Generate Low-Stock Alert; Email / SMS Gateway; Notification class

Store	**Class**
D1 User	User
D2 Supplier	Supplier
D3 Product	Product
D4 Stock Record	StockRecord

D5 Sales Order	SalesOrder
D6 Forecast	Forecast
D7 Purchase Order	PurchaseOrder
D8 Notification	Notification

DFD Level 2

Two Level 1 processes are decomposed: 3.0 Process Sales Order and 4.0 Manage Inventory Operations. Both follow the step order of their activity-diagram swimlanes, and each keeps exactly the external flows its parent has at Level 1, so the levels stay balanced.

Process 3.0 – Process Sales Order

Follows the Sales Personnel lane: Create Sales Order → Scan / Select Product → Check Stock Availability → (No: Show Out of Stock Message | Yes: Process Payment) → Generate Invoice → Update Sales Order. Online orders enter at 3.1 through the Marketplace Webhook, and 3.6 writes the order to D5 and deducts stock in D4 (SalesOrder.create_order(), StockRecord.deduct_stock()).

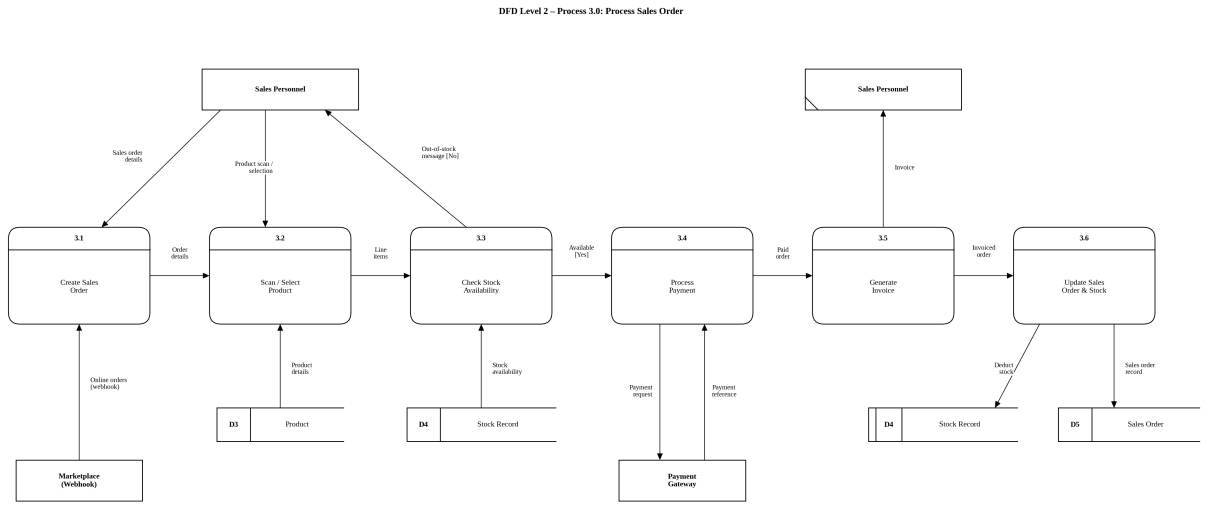


Figure 3: DFD Level 2 – Process 3.0 Process Sales Order

Process 4.0 – Manage Inventory Operations

Follows the Warehouse Staff lane: Select Inventory Operation → Use Voice Command? (Yes: Speak Command → Speech-to-Text Processing | No: Use Standard UI) → Interpret Command → Confirm Operation → Update Stock Record. 4.2 calls the Speech-to-Text service, 4.4 plays the confirmation back through the Text-to-Speech service, and 4.5 writes to D4 (WarehouseStaff.capture_voice_command(), StockRecord.update_stock()).

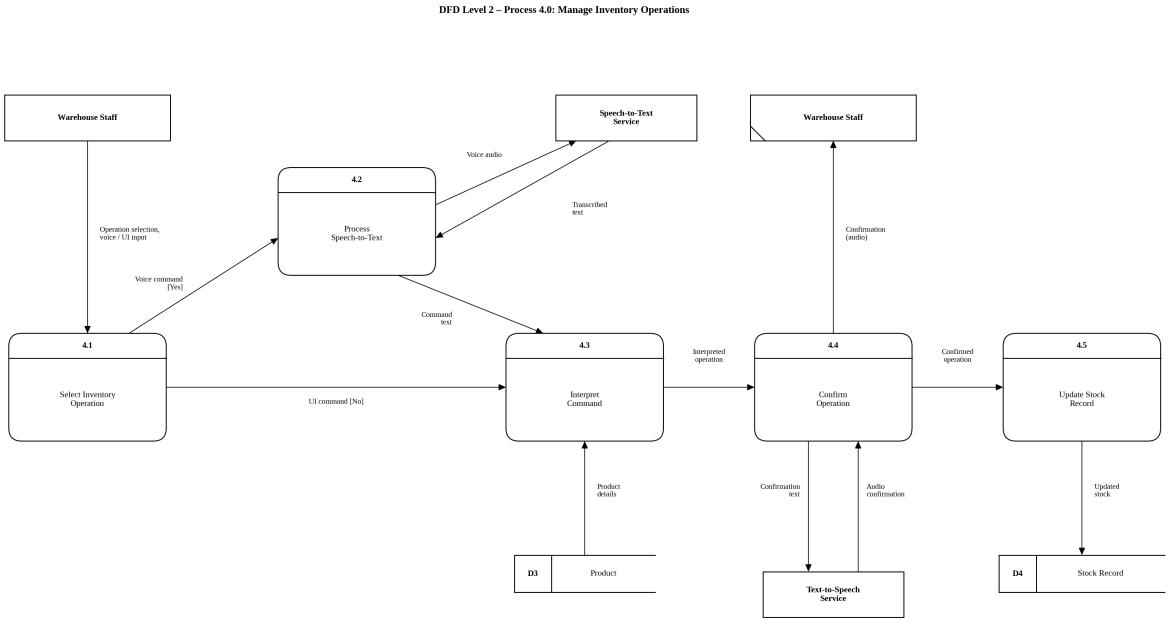


Figure 4: DFD Level 2 – Process 4.0 Manage Inventory Operations

Conclusion

The Data Flow Diagrams for the Inventory and Stock Management System were successfully drawn at Level 0, Level 1 and Level 2 using Gane-Sarson notation. The context diagram fixed the system boundary against eleven external entities; Level 1 split the system into seven processes and eight data stores that map one-to-one onto the activity-diagram swimlanes and the class diagram's classes; and Level 2 refined Process Sales Order and Manage Inventory Operations into the exact step sequences of their swimlanes, including the stock-availability and voice-command decisions.

Flows were kept balanced between levels, so every input and output of a parent process reappears in its child diagram. Together, the diagrams show how data moves from users and external services through processing into persistent storage, and they give a clear basis for designing the database and modules of the system.